EE115 Analog Circuits
模拟电路

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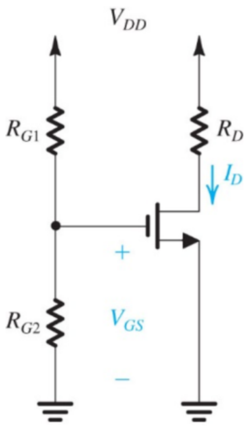
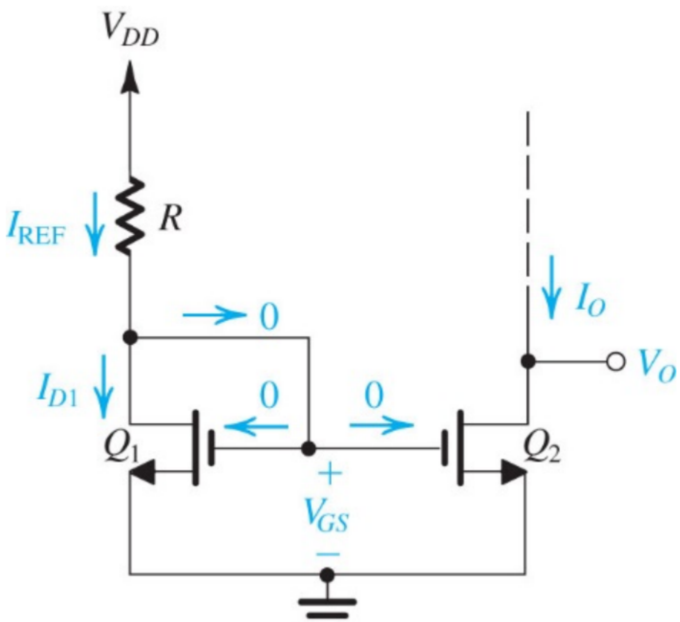
Outline

- Building blocks of IC amplifier
- SEDTRA/SMITH book page 525-556；

Review:Current mirror





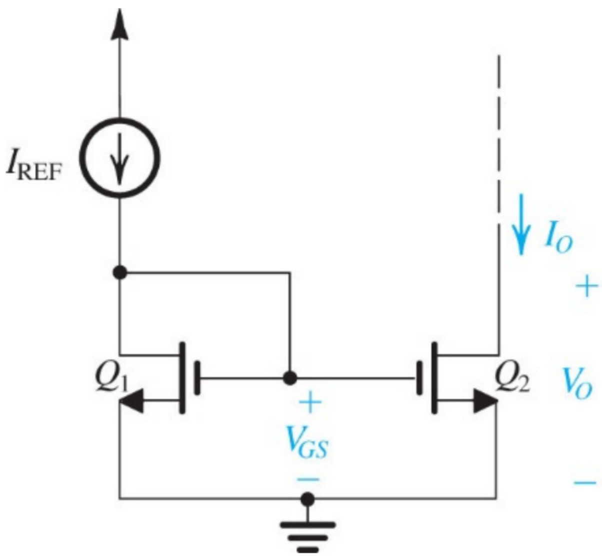
- Neglecting channel length modulation

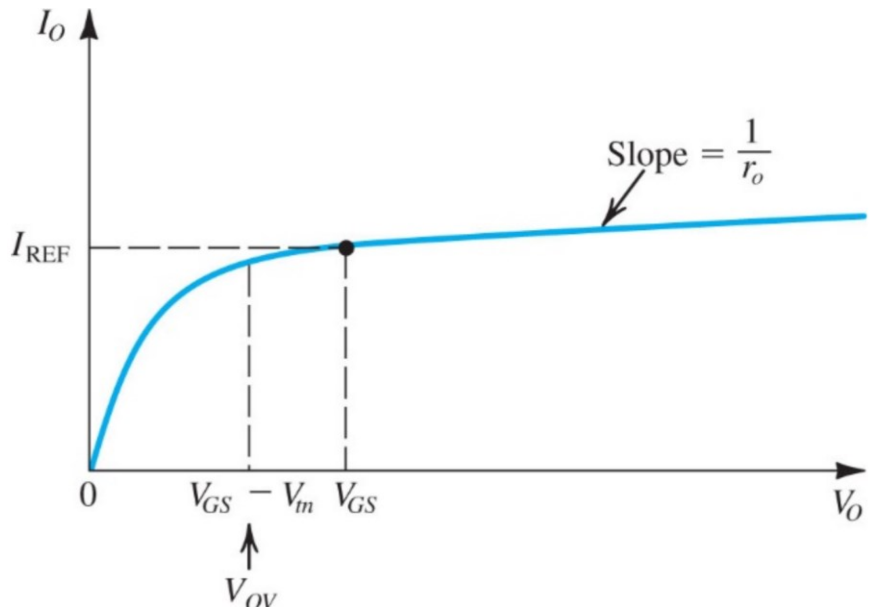
$$I_{D1} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_1 (V_{GS} - V_{tn})^2$$

$$I_O = I_{D2} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_2 (V_{GS} - V_{tn})^2$$

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

Review: Current Mirrors with reference source





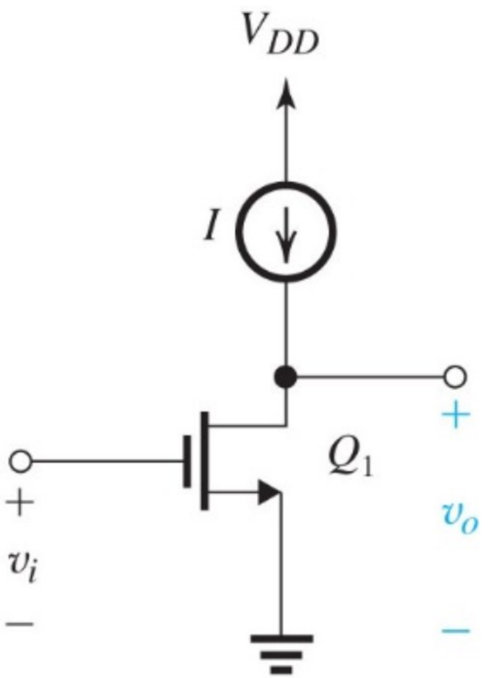
- Neglecting channel length modulation

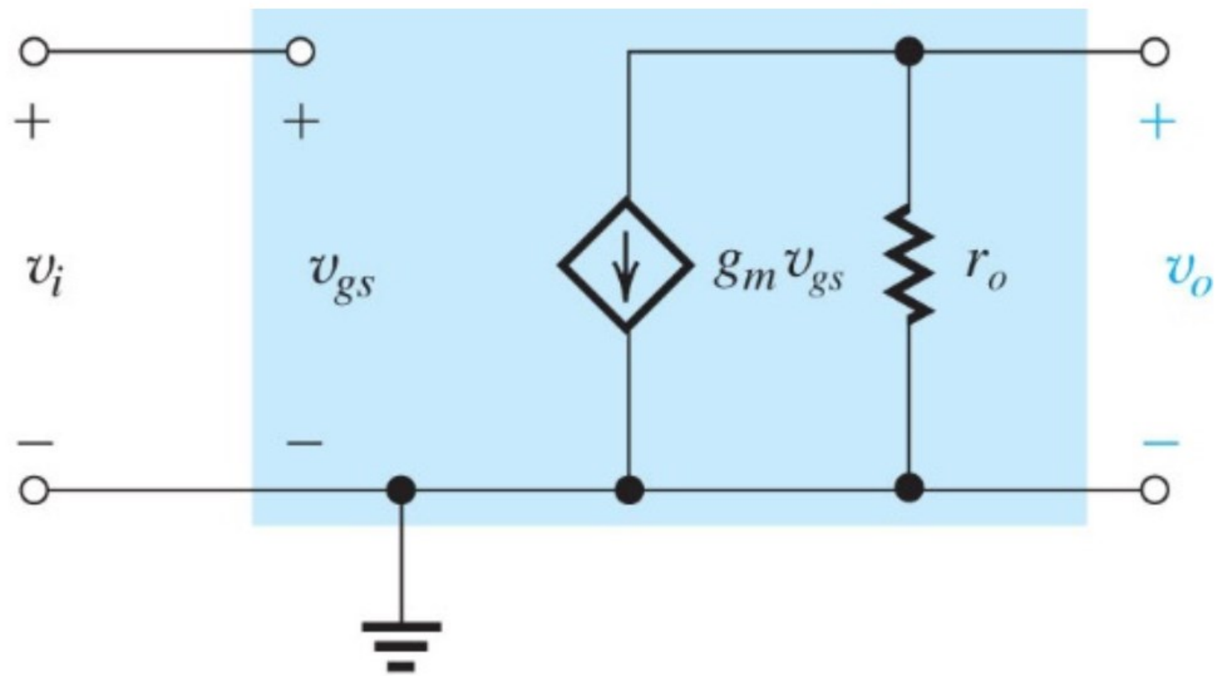
$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

- If considering channel length modulation

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF} \left(1 + \frac{V_O - V_{GS}}{V_{A2}}\right)$$

Review: Basic Gain Cells with Ideal Current Source Load





$$g_m = \frac{I_D}{V_{OV}/2}$$

$$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D}$$

$$r_o = \frac{V_A}{I_D} = \frac{V_A' L}{I_D}$$

$$A_0 = \frac{V_A}{V_{OV}/2} = \frac{2V_A' L}{V_{OV}}$$

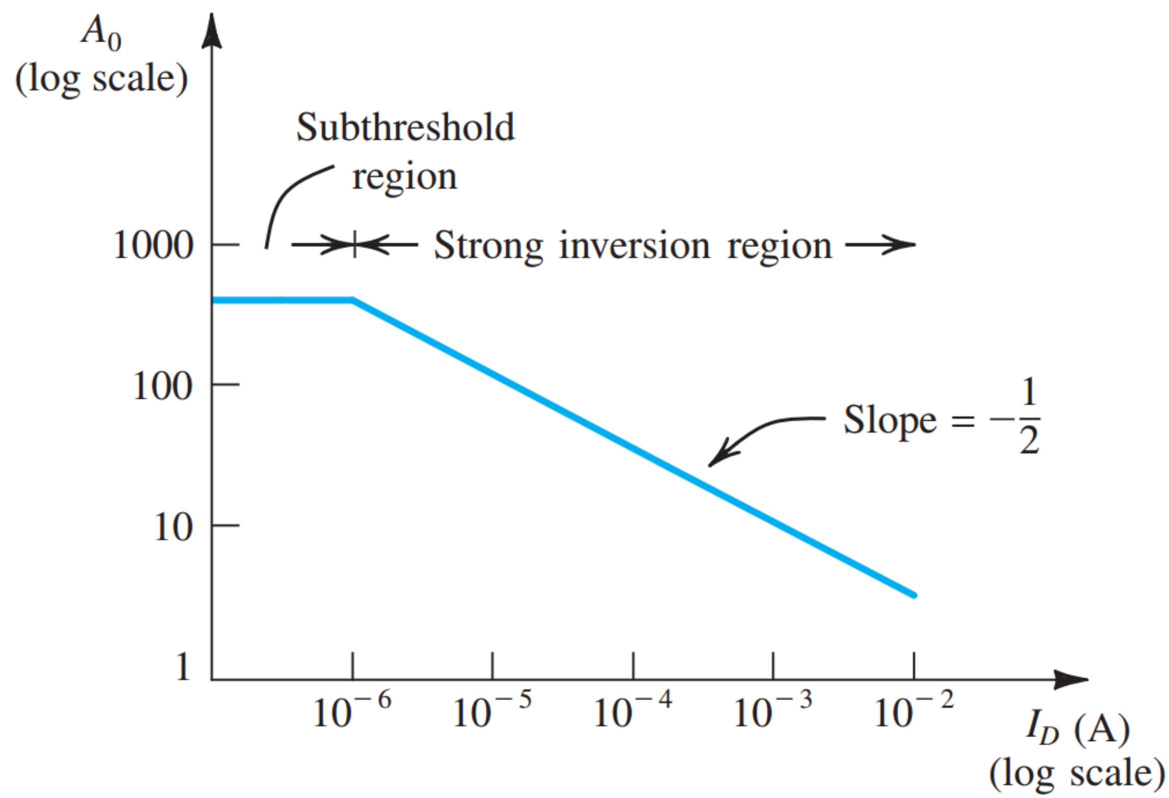
$$R_{\text{in}} = \infty$$

Intrinsic gain: $A_{vo} = -g_m r_o$

$$R_o = r_o$$

Review: The Intrinsic Gain $g_m r_o$

- MOSFET



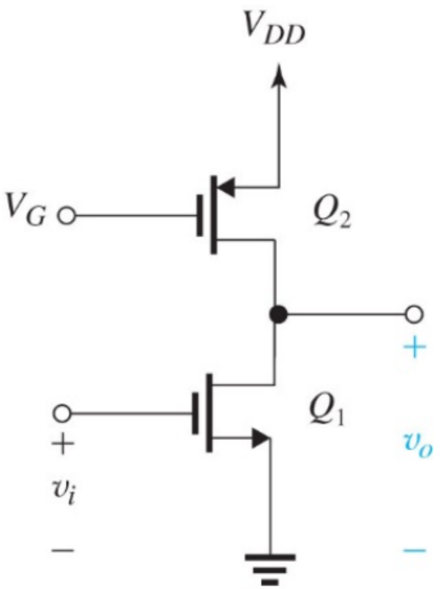
$$A_0 = \frac{V_A' \sqrt{2(\mu_n C_{ox})(WL)}}{\sqrt{I_D}}$$

$$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D} \qquad r_o = \frac{|V_A|}{I_D}$$

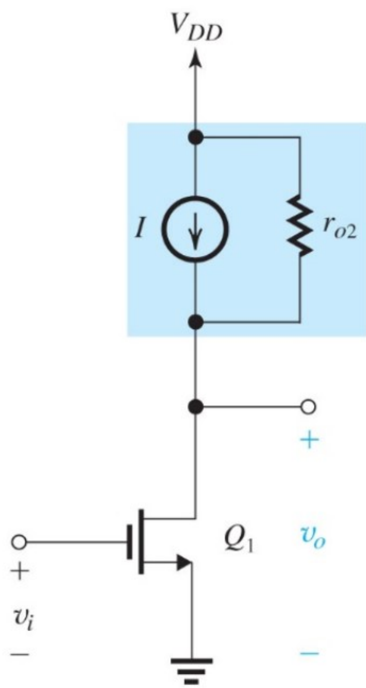
$$Vov = \sqrt{2I_D/(k_n'(W/L))}$$

Smaller current gives higher gain, but paid by smaller g_m and bandwidth.

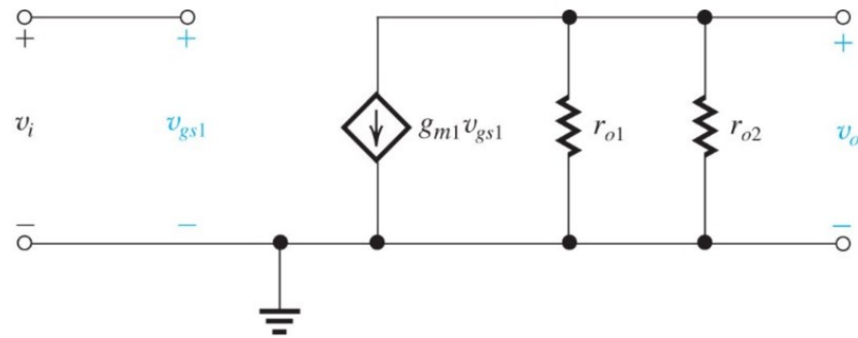
Review: Output Resistance of Current Source Load



(a)



(b)



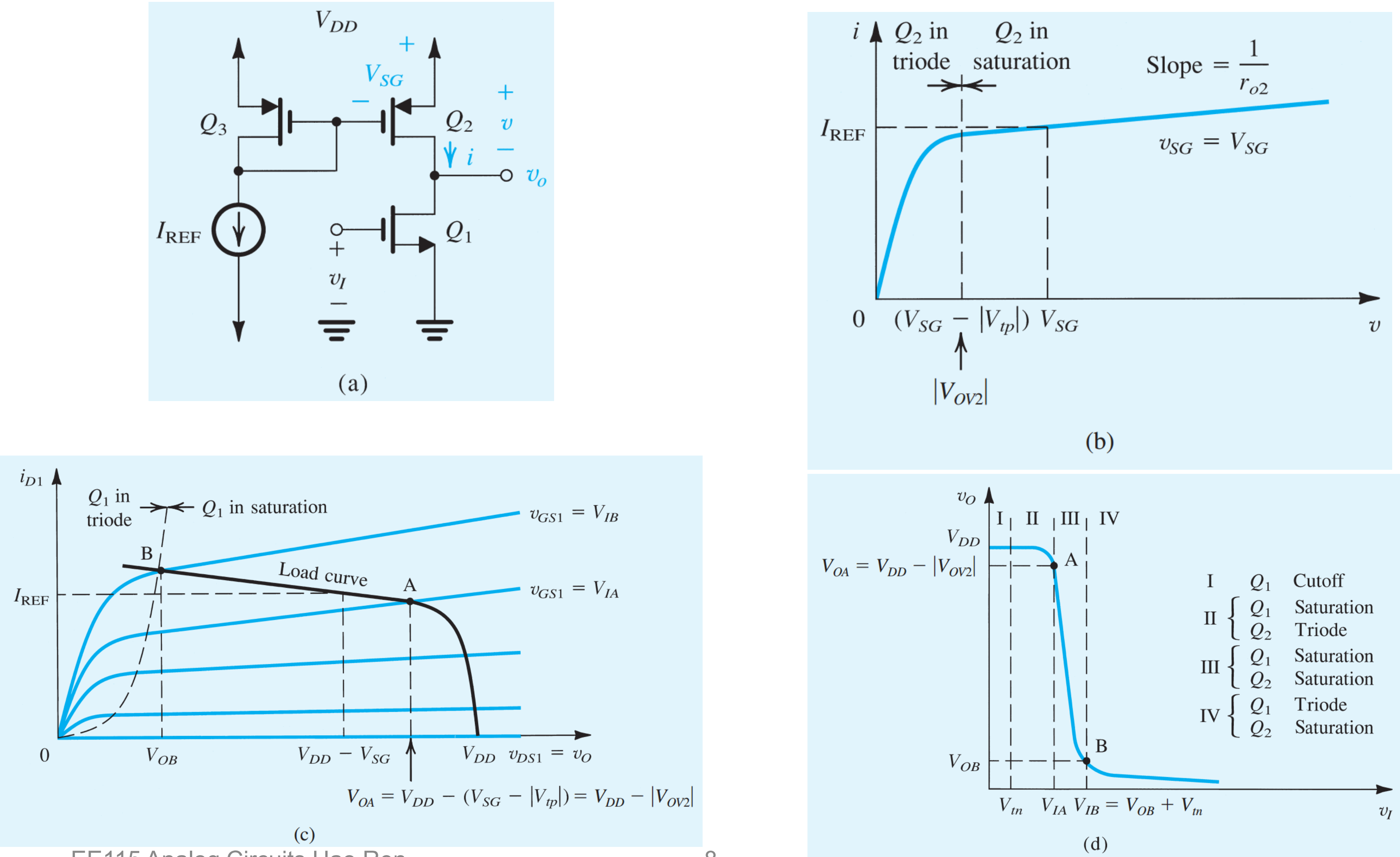
(c)

$$r_{o2} = \frac{|V_{A2}|}{I}$$

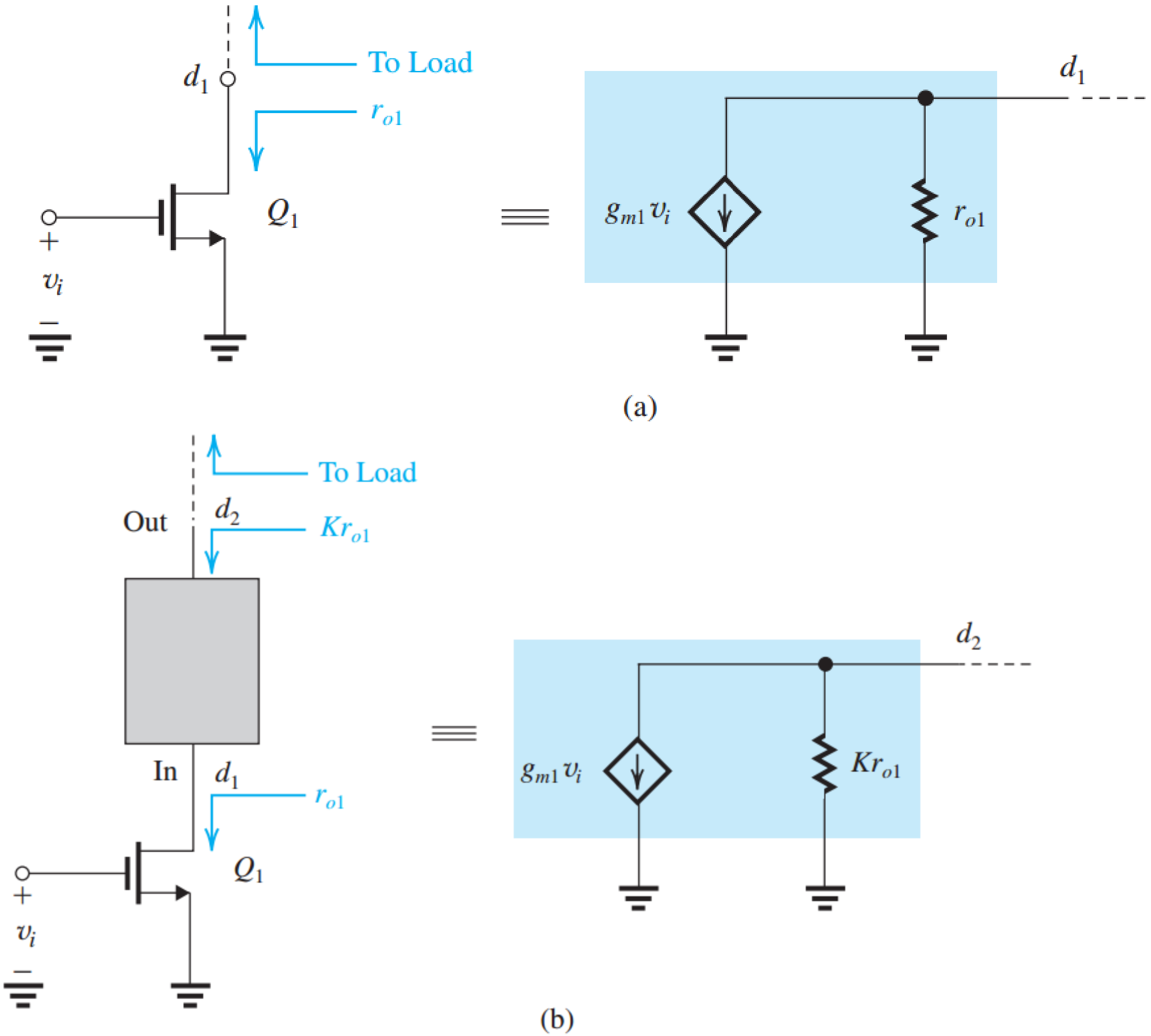
$$I = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_2 \left[V_{DD} - V_G - |V_{tp}|\right]^2$$

$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \| r_{o2})$$

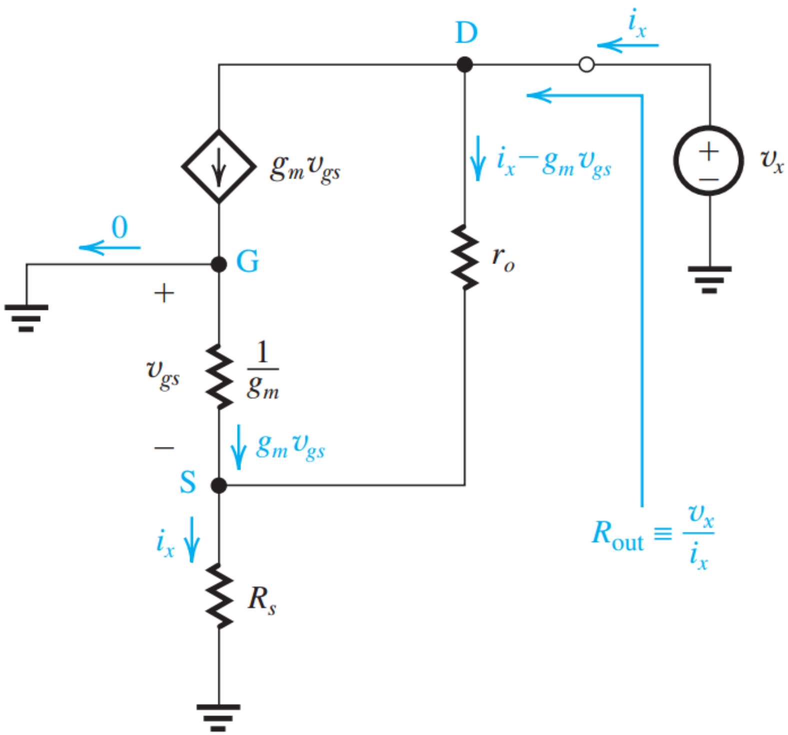
Graphical Solution of Voltage Transfer Curve



How To Increase Voltage Gain?



Continued 1



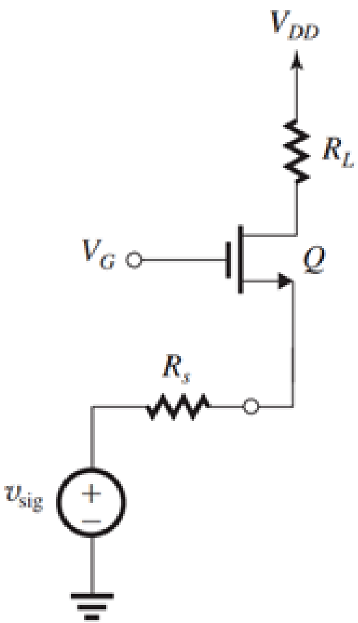
$$v_x = (i_x - g_m v_{gs})r_o + i_x R_s$$

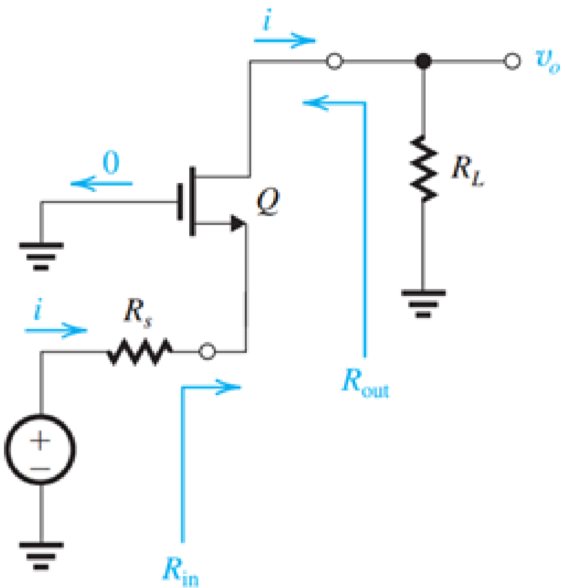
$$v_{gs} = -i_x R_s$$

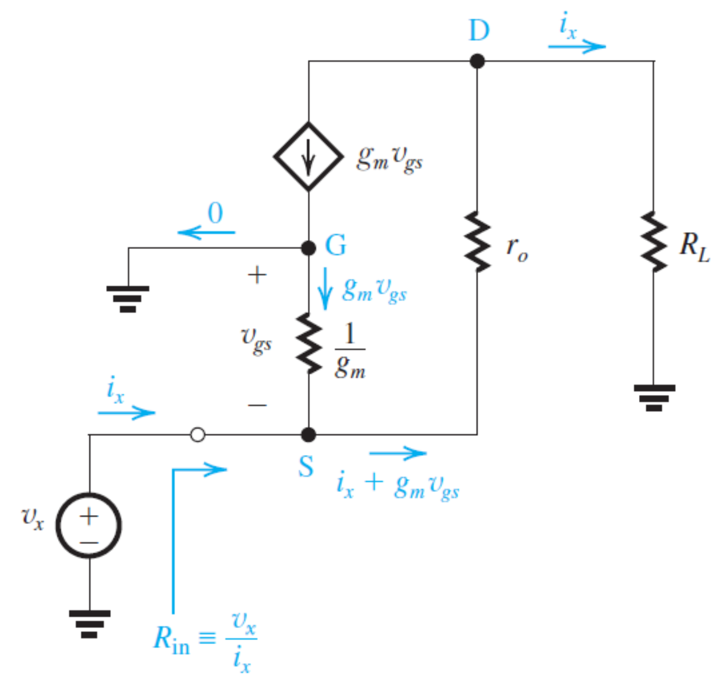
$$\begin{aligned} R_{out} &= r_o + R_s + g_m r_o R_s \\ &= r_o + (1 + g_m r_o) R_s \\ &\simeq (g_m r_o) R_s \end{aligned}$$

The source impedance is transformed to the output by multiplying the intrinsic gain.

Impedance Transformation of Common Gate Amplifier





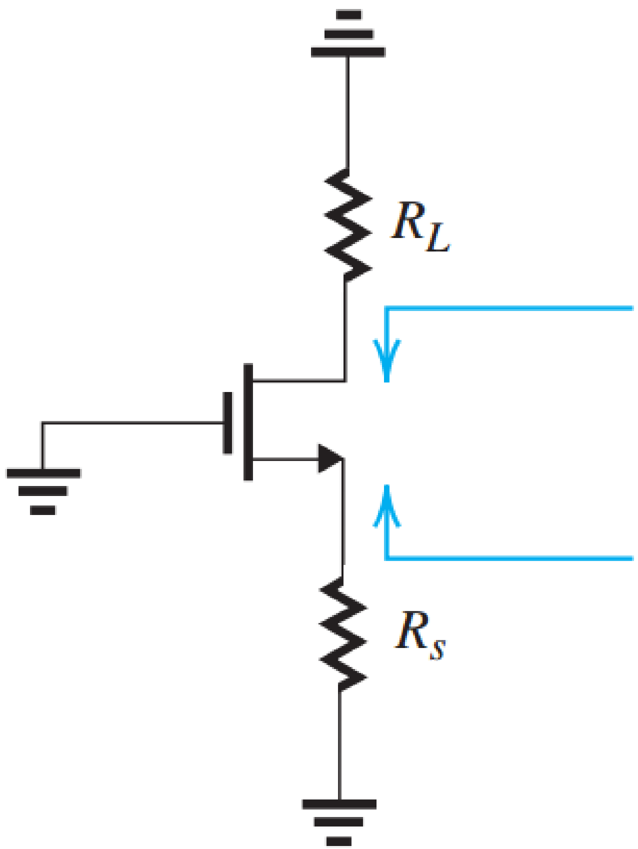


$$v_x = (i_x + g_m v_{gs}) r_o + i_x R_L$$

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o} \simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

The load impedance is transformed to the input by dividing by the intrinsic gain.

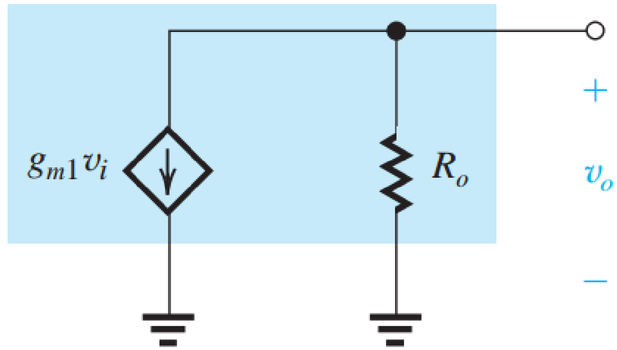
Continued 2



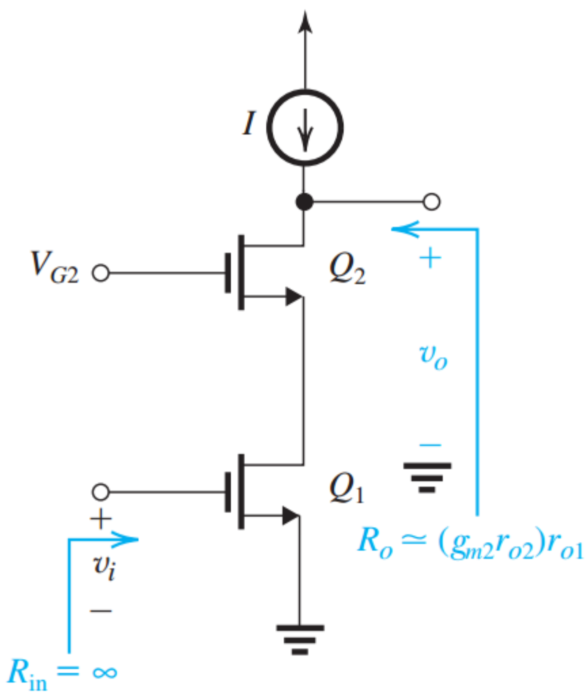
$$R_{\text{out}} = r_o + R_s + g_m r_o R_s$$
$$\simeq r_o + (g_m r_o) R_s$$

$$R_{\text{in}} = \frac{r_o + R_L}{1 + g_m r_o}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

MOS Cascode Amplifier



(b)



(a)

$$A_{vo} \equiv \frac{v_o}{v_i} = -g_{m1}R_o$$

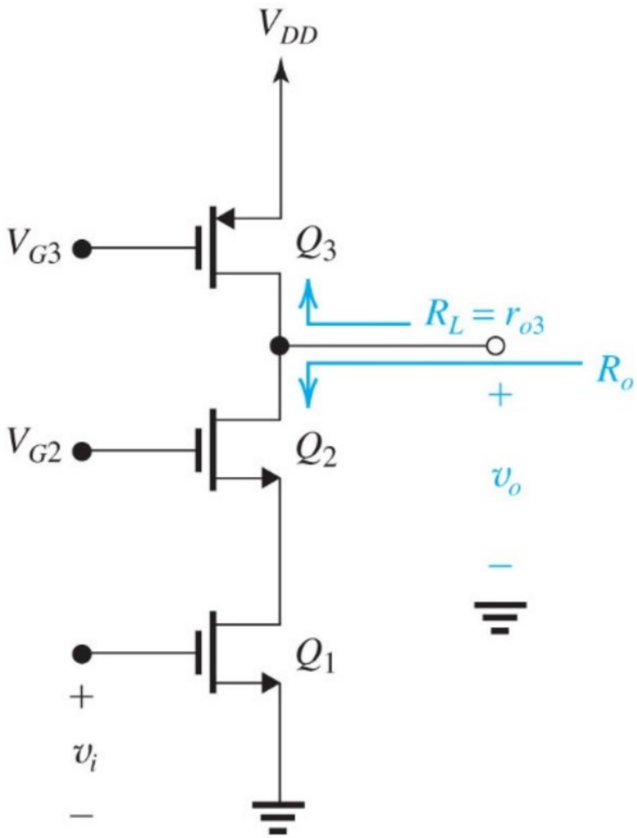
$$R_o \simeq (g_{m2}\,r_{o2})\,r_{o1}$$

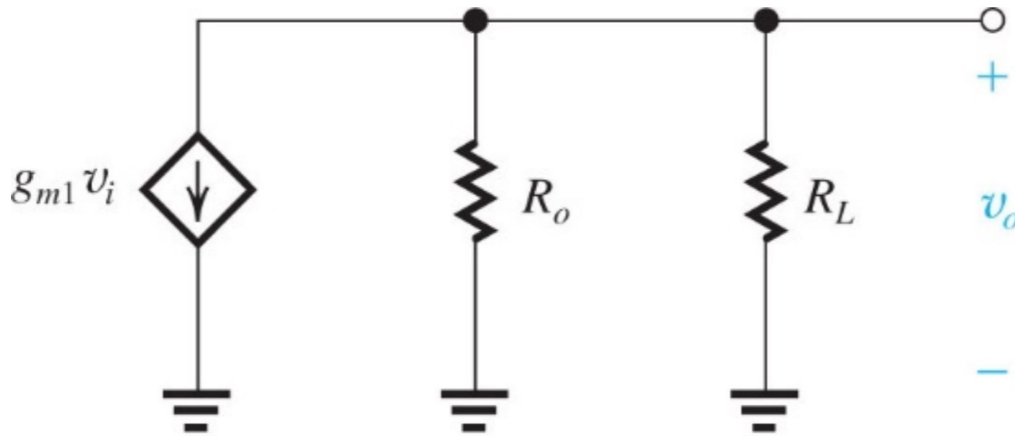
$$A_{vo} = -(g_{m1}\,r_{o1})(g_{m2}\,r_{o2})$$

$$A_{vo} = -(g_m\,r_o)^2$$

$$= -A_0^2$$

Cascode Amplifier with Simple Active Load

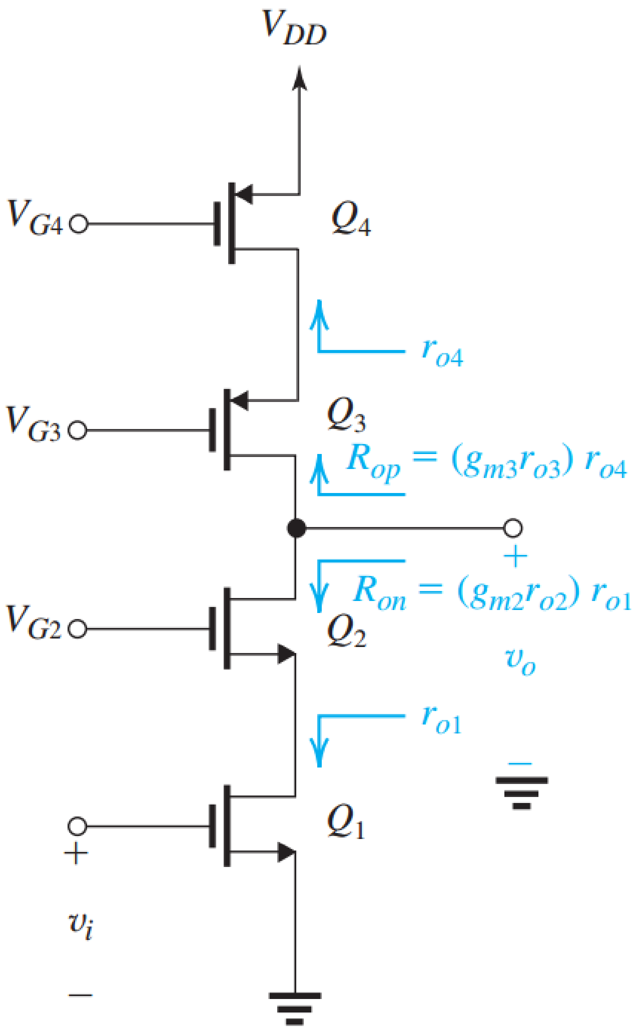




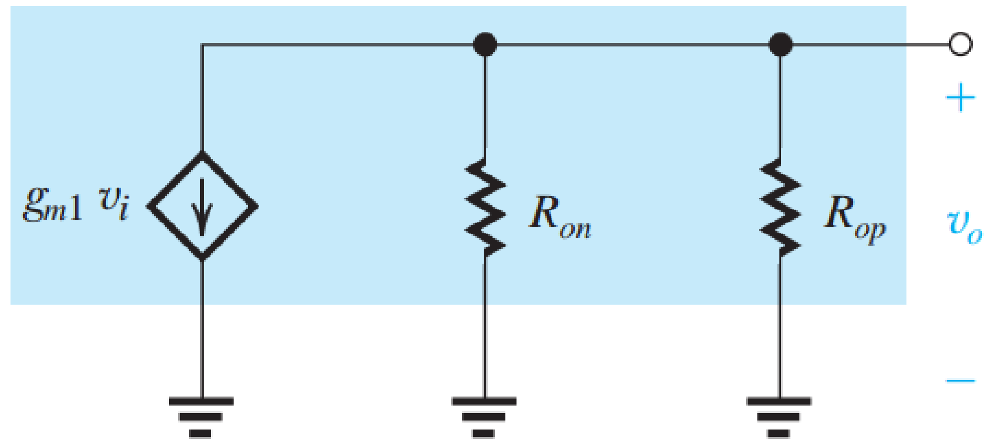
(b)

$$
\begin{aligned}
A_v &= -g_{m1}(R_o \| R_L) \\
&= -g_{m1}(g_{m2} r_{o2} r_{o1} \| r_{o3}) \\
&\simeq -g_{m1} r_{o3}
\end{aligned}
$$

Cascode Amplifier with Cascode Current-Source Load



(a)



(b)

$$A_v = -g_{m1}\{[(g_{m2}r_{o2})r_{o1}] \| [(g_{m3}r_{o3})r_{o4}]\}$$

$$A_v = -\frac{1}{2}(g_m r_o)^2 = -\frac{1}{2}A_0^2$$

Homeworkx-due April xx   11p.m.

SEDRA/SMITH Book chapter x problem xx